Household Credit, Risk Premia and Consumption

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- Risk premia are an important source of business cycle fluctuations
  Gilchrist Zakrajšek (12), Pflueger Siriwardane Sunderam (20)

- Dominant theoretical view: price of capital is key transmission channel
  Gourio (12), Brunnermeier Sannikov (14), Caballero Simsek (20)

- But, some tension with broader macro & empirical macro-finance

  - Households with low liquid wealth are important for consumption fluctuations
    Kaplan Violante (18), Aguiar Bils Boar (25)

  - Household debt also predicts consumption fluctuations
    Mian Sufi (18), Mian Sufi Verner (17a, 17b, 20)

This paper: household credit supply is the key link between finance and macro

Overview

Quantitative NK model capital and household debt, shocks to price of risk

- Three channels: household credit, capital wealth effects, investment
- Analytical insights
  - Strength of channels identified by cross-sectional cons. responses
- Empirical evidence from the Consumer Expenditure Survey
  - Consumption growth strongly co-moves with risk premia for borrowers
  - Little evidence for wealth effects or investment channel
- Quantification matching SCF portfolios & time series moments
  - Credit supply accounts for most of the cons. response to fin. shocks
  - Countercyclical macro-pru complementary when applied across asset markets

Related literature

- Risk premia in business cycle models

  Tallarini (00), Gourio (12), Brunnermeier Sannikov (14), Basu Bundick (17), Caballero Simsek (20), Elenev Landvoigt Van Nieuwerburgh (21), Kekre Lenel (22)

  Here: risk premia fluctuations affect price of household credit

- Heterogeneity in macro

  Cúrdia Woodford (16), Kaplan Moll Violante (18), Auclert Rognlie Straub (20), Aguiar Bils Boar (25), Lewis Melcangi Pilossoph (26), Faccini Lee Luetticke Ravn Renkin (26)

  Here: Euler eq. holds, changing credit conditions rather than incomes, risk premia

- Empirical studies of credit supply and household demand

  Gross Souleles (02), Benmelech Meisenzahl Ramcharan (17), Mian Sufi Verner (17a, 17b, 20), Agarwal Chomsisengphet Mahoney Stroebel (18), Aydin (22), Garber Mian Ponticelli Sufi (24)

  Here: novel moments; structural model consistent with this evidence

- Consumption based asset pricing

  Hansen Singleton (82, 83), Mankiw Zeldes (91), Vissing-Jørgensen (02), Yogo (04), Parker Julliard (05), Malloy Moskowitz Vissing-Jørgensen (09), Di Tella Hébert Kurlat Wang (26)

  Here: a large share of households respond to risk prices, but do not price risk

Model

Model

- Two household types: investors i, debtors d, discount factors $\beta^i > \beta^d$
  - Utility over consumption and labour, $u(c_t, l_t) = (c_t \Phi(l_t))^{1-1/\psi}$
  - Households die at rate ξ; investors are born with $\bar{\eta}$ share of wealth
- Financial markets: capital, risky bonds, safe bonds
  - Portfolio choice delegated to an intermediary that solves

$$\max_{\{\omega^j\}} \mathbb{E}_t \left[\frac{n_{t+1}^{1-\gamma_t}}{1-\gamma_t} \right] \quad \text{s.t.} \quad \overbrace{n_{t+1}}^{\text{next period wealth}} = \sum_j \omega^j R_{t+1}^j n_t, \quad \sum_j \omega^j = 1$$

  - One period risky bonds: price q_t^b and exogenous default, rebated lump sum
  - Monetary policy follows a Taylor rule, $(1 + i_t) = (1 + \bar{i}) \left(\frac{P_t}{P_{t-1}} \right)^{\phi}$

Model

- Supply-side
  - Union sets wages subject to Rotemberg costs, allocates $l_t^j = \bar{l}^j l_t$, $\int \bar{l}^j = 1$
  - Cobb-Douglas production $y_t = (\exp(z_t) l_t)^{1-\alpha} k_{t-1}^{\alpha}$
  - Investment subject to adjustment costs $\implies q_t^k = (k_t / k_{t-1})^{\chi^k}$
- Driving forces
  - Intermediary risk appetite, $\gamma_t = (1 - \rho^\gamma)\bar{\gamma} + \rho^\gamma \gamma_{t-1} + \sigma^\gamma \epsilon_t^\gamma$ Leverage equivalence
  - Productivity follows a random walk with disaster risk, $z_t = z_{t-1} + \sigma^z \epsilon_t^z + \varphi_t$
- **Equilibrium:** agent optimisation; goods, labour, asset markets clear Full definition

Analytical insights

Simplifying assumptions, relaxed in quantitative model

- Two periods $t \in \{0, 1\}$
- Log utility $\psi = 1$, exogenous labour $\Phi(l_t) = 1$
- Lognormal shocks:
  - Default $\epsilon_1^b \sim \mathcal{N}(-\frac{1}{2}\sigma_b^2, \sigma_b^2)$
  - Productivity $\epsilon_1^z \sim \mathcal{N}(-\frac{1}{2}\sigma_z^2, \sigma_z^2)$, $\epsilon_1^z \perp\!\!\!\perp \epsilon_1^b$.
- Fixed nom. wage W_0. Fixed labour endowment at $t = 1$, $\bar{w}$. No investor labour.
- No initial nominal positions, $b_{-1}^j = m_{-1}^j = 0$, investor holds capital stock k_{-1}
- Policy: $d \log(1 + i_0)/d\epsilon_0^\gamma = 0$, $P_1 = P_0$.

How do changes in risk premia affect consumption and investment?

- Risk premia are linear in intermediary risk aversion

$$\mathrm{rp}_0^b = E\log(1+r_1^b) - \log(1+r_0^f) + \frac{1}{2}\sigma_b^2 = \sigma_b^2\gamma_0^i\frac{-a_0^d}{a_0^i},$$

$$\mathrm{rp}_0^k = E\log(1+r_1^k) - \log(1+r_0^f) + \frac{1}{2}\sigma_z^2 = \sigma_z^2\gamma_0^i\frac{a_0^i + a_0^d}{a_0^i}, \quad a_0^j \text{ is type } j \text{ saving}$$

- Consumption is a constant share of total lifetime wealth

$$c_0^d = (1-\beta^d)\left(w_0 l_0 + q_0^b \bar{w}\right),$$
$$c_0^i = \left(1-\beta^i\right)(\Pi_0 + q_0^k)k_{-1}.$$

- Investment is decreasing in the capital return

$$E\log(1+r_1^k) = -(1-\alpha+\chi^k)\log(k_0) - \frac{1}{2}\sigma_z^2 + \chi^k\log(k_{-1}).$$

Response to a risk aversion shock

- In response to a fall in risk aversion, investment rises and borrowing rates fall

$$\frac{dk_0}{d\epsilon_0^\gamma} = -\frac{k_0}{1-\alpha+\chi^k}\frac{d\mathrm{rp}_0^k}{d\epsilon_0^\gamma}, \quad \frac{dq_0^b}{d\epsilon_0^\gamma} = -q_0^b\frac{d\mathrm{rp}_0^b}{d\epsilon_0^\gamma},$$

- The change in aggregate consumption is given by

$$\frac{dC}{d\epsilon_0^\gamma} = -\frac{1}{\tilde{\beta}}\overbrace{(1-\tilde{\beta})\frac{(1+\chi^k)q_0^k k_0 - \chi^k q_0^k k_{-1}}{1-\alpha+\chi^k}\frac{d\mathrm{rp}_0^k}{d\epsilon_0^\gamma}}^{\equiv -\mathcal{C}^x,\ \text{Investment effect}} - \frac{1}{\tilde{\beta}}\overbrace{(1-\beta^i)\frac{\chi^k q_0^k k_{-1}}{1-\alpha+\chi^k}\frac{d\mathrm{rp}_0^k}{d\epsilon_0^\gamma}}^{\equiv -\mathcal{C}^k,\ \text{Capital wealth effect}}$$
$$- \frac{1}{\tilde{\beta}}\underbrace{(1-\beta^d)q_0^b\bar{w}\frac{d\mathrm{rp}_0^b}{d\epsilon_0^\gamma}}_{\equiv -\mathcal{C}^b,\ \text{Credit supply effect}}.$$

$\rightarrow$ all three channels link risk premia and consumption

Identification in the cross-section

- Cons. response for households without debt or capital (measure zero, index n) is

$$\overbrace{\frac{d \log c_0^n}{d\epsilon_0^\gamma}}^{\text{Cons. response to risk price shock}} = \overbrace{\frac{d \log y_0}{d\epsilon_0^\gamma}}^{\text{GE (incl. investment)}}$$

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- The change in consumption for debtors adds credit supply channel

$$\frac{d \log c_0^d}{d\epsilon_0^\gamma} = (1-\theta^b)\frac{d \log y_0}{d\epsilon_0^\gamma} + \overbrace{\theta^b \frac{dq_0^b}{d\epsilon_0^\gamma}}^{\mathcal{C}^b/c_0^d}, \qquad \theta^b \equiv \frac{q_0^b \bar{w}}{w_0 l_0 + q_0^b \bar{w}}$$

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- The change in consumption for investors adds capital wealth channel

$$\frac{d \log c_0^i}{d\epsilon_0^\gamma} = (1-\theta^k)\frac{d \log y_0}{d\epsilon_0^\gamma} + \overbrace{\theta^k \frac{dq_0^k}{d\epsilon_0^\gamma}}^{C^k/c_0^i}, \qquad \theta^k \equiv \frac{q_0^k}{\Pi_0 + q_0^k}$$

Identification in the cross-section

- $\frac{d \log c_0^n}{d\epsilon_0^\gamma} = \frac{d \log y_0}{d\epsilon_0^\gamma}$ is an upper bound on the investment effect

- The credit supply effect is bounded by

$$\theta^b \frac{dq_0^b}{d\epsilon_0^\gamma} \in \left(\Delta^d,\ \Delta^d + \frac{d \log y_0}{d\epsilon_0^\gamma} \right) \qquad \Delta^d \equiv \frac{d \log c_0^d}{d\epsilon_0^\gamma} - \frac{d \log c_0^n}{d\epsilon_0^\gamma}$$

- The capital wealth effect is bounded by

$$\theta^k \frac{dq_0^k}{d\epsilon_0^\gamma} \in \left(\Delta^i,\ \Delta^i + \frac{d \log y_0}{d\epsilon_0^\gamma} \right) \qquad \Delta^i \equiv \frac{d \log c_0^i}{d\epsilon_0^\gamma} - \frac{d \log c_0^n}{d\epsilon_0^\gamma}$$

- The strength of the channels is identified in hh-level cons. and balance sheet data

Empirical evidence

Data

- US data, 1994 to 2025
- Household consumption from the Consumer Expenditure Survey (CEX)
  - Real semiannual non-durables and services cons. growth (also total inc. durables)
  - Balance sheet information: CEX shares SCF portfolios
    - Direct holdings of stocks, bonds or mutual funds
    - Consumer credit (cards, auto, personal, . . .) & HELOCs/HELOANs
    - Alternative specifications with mortgages, retirement accounts
  - Monthly repeated cross-sections
- Risky asset prices
  - Excess bond premium from Gilchrist Zakrajšek (12), six month average
  - Robust to other risky asset price measures

    Bekaert, Engstrom and Xu (2022) Campbell and Shiller (1988)

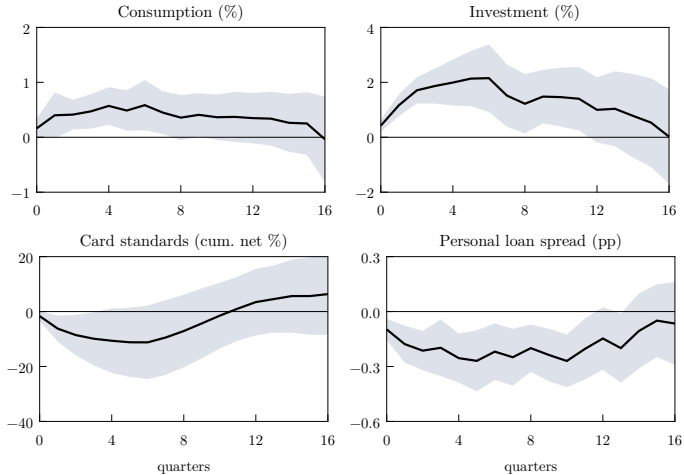
Local projections

- I project personal loan spreads, consumption, debt and investment on the EBP

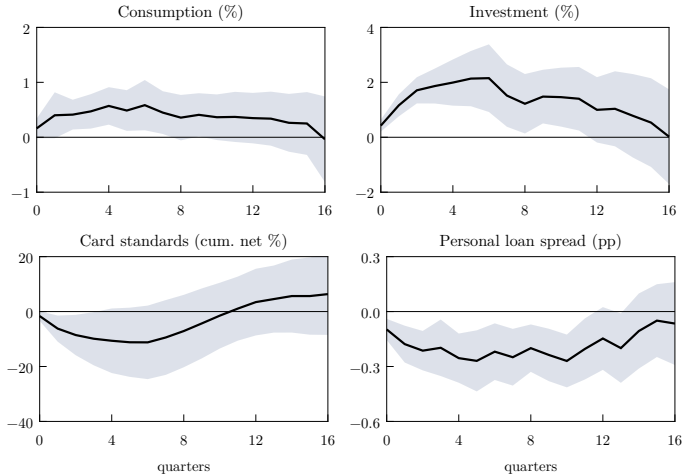
$$y^i_{t+h} = \alpha_h + \theta_h EBP_t + \sum_{l=1}^{p} \delta'_{h,l} w_{t-l} + \epsilon_{h,t}$$

- $w_t = (r'_t, EBP_t, y'_t)$ where r'_t is inflation, term spread, 2y yield
- y_t is
  - Real log per capita non-durables and services personal cons. expenditures
  - Revolving consumer credit to trend GDP
  - Real log private fixed investment
  - 2-year personal loan rate to 2-year Treasury spread

Impulse response to a one standard deviation fall in the EBP

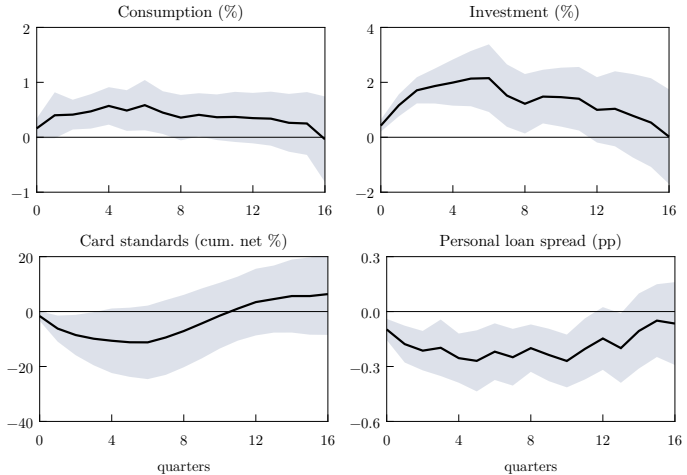


Impulse response to a one standard deviation fall in the EBP



- Standard story: investment boom feeds back into consumption . . .

Impulse response to a one standard deviation fall in the EBP



- Standard story: investment boom feeds back into consumption ...
  ... but cannot disentangle role of household credit supply from time series alone

Empirical strategy

- Cross-sectional diff. in consumption risk price correlation identify channels

$$\Delta \log c_t^i = \alpha + \sum_j \beta^j EBP_t \times \mathbf{1}[i \in S^j] + \sum_j \gamma^j EBP_{t-6} \times \mathbf{1}[i \in S^j] + \delta' X_t^i + \epsilon_t^i$$

where $S^j \in \{\text{cons. cred. \& sec; cons. cred. \& no sec; sec \& no cons. cred.; all}\}$

- Model mapping:

$$\begin{aligned}
\frac{d \log y_0}{d\epsilon_0^\gamma} \text{ (GE)} &= \beta^{\text{all}} && < 0 \\
\theta^b \frac{dq_0^b}{d\epsilon_0^\gamma} \text{ (credit supply)} &\in \left(\beta^{\text{cc \& no sec}},\ \beta^{\text{cc \& no sec}} + \frac{d \log y_0}{d\epsilon_0^\gamma}\right) && < 0 \\
\theta^k \frac{dq_0^k}{d\epsilon_0^\gamma} \text{ (capital wealth)} &\in \left(\beta^{\text{sec \& no cc}},\ \beta^{\text{sec \& no cc}} + \frac{d \log y_0}{d\epsilon_0^\gamma}\right) && < 0
\end{aligned}$$

Empirical strategy

- Threats to identification:
  1. Different cyclical income exposures
     → Within occupation, income, age estimates (group × time fixed effects)
     → Placebo test: wages and salaries, unemployment rate (w/ EBP control)
     Wages Unemployment
  2. Different consumption baskets
     → Food consumption only Regression results
  3. Safe rates
     → 5-year TIPS yield (w/ EBP control) Regression results
- Other specifications: Mortgages Retirement accounts excl. recessions

Consumption and the EBP in the cross-section

	Nondurables and services	
	(1)	(2)
EBP	0.225 (0.774)	—
Sec., no cons. credit × EBP	-1.001 (0.901)	-0.569 (0.755)
Cons. credit, no sec. × EBP	-1.746*** (0.515)	-1.772*** (0.491)
Cons. credit & sec. × EBP	-2.084 (1.271)	-2.159** (0.961)
Fixed effects	None	Demog. × date
Observations	122,407	122,407

- No response from baseline group → weak investment/GE channel

Consumption and the EBP in the cross-section

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- Weak response from securities holders → small wealth effects

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- Strong response from households with consumer credit

Consumption and the EBP in the cross-section

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Sec., no cons. credit × EBP	-1.001 (0.901)	-0.569 (0.755)
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- Need a model with heterogeneous contemporaneous correlation from borrowers
- Consistent with credit supply channel, not the others

Quantitative model

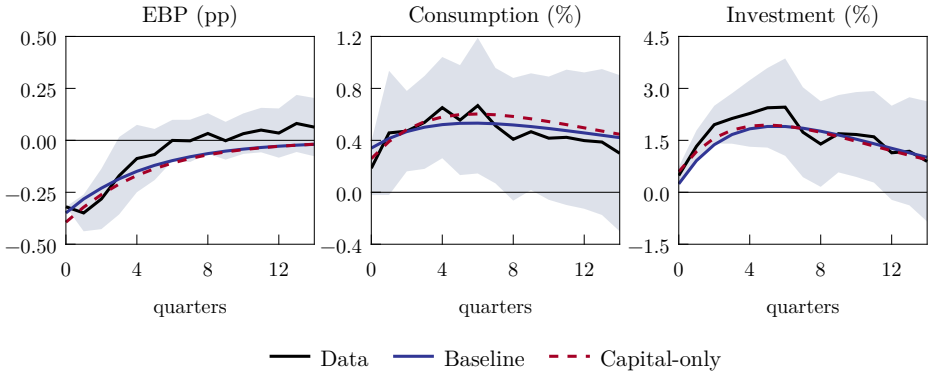
Additional model features, calibration and solution

- Additional quantitative features
  - Counter-cyclical default rates
  - Minimum capital holdings $k_t^j \geq \bar{k} n_t^j$
- Target micro moments (SCF portfolios, labour income, population size)

  and macro moments (equity premium, credit spreads, consumption, investment)

  SCF shares | External calibration | Internal calibration
- Global solution over five states: $\eta_t, w_{t-1}, k_{t-1}, \gamma_t, \eta_t^f$

Model validation: impulse responses



- Lower intermediary risk aversion → consumption & investment boom

Model validation: consumption growth regressions

- Capital only model (rep agent) calibrated to same set of moments Calibration

	Data	Baseline	Capital-only
$\hat{\beta}$: $\Delta \log c_t = \alpha + \beta EBP_t + \gamma EBP_{t-1} + \epsilon_t$			
Aggregate	-2.4 (1.1)	-2.4	-2.1
Investors	-1.1 (1.2)	-1.3	-2.1
Debtors	-2.3 (0.9)	-2.9	–

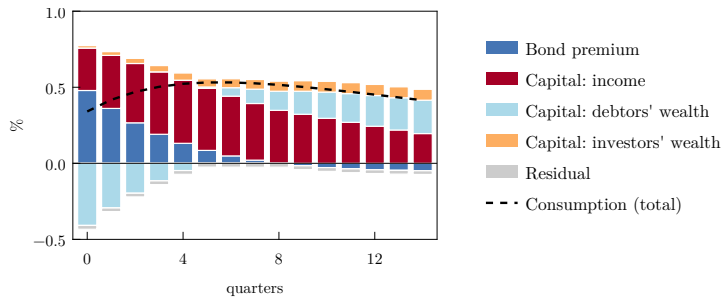
- Capital only model does not deliver enough comovement in aggregate
- Baseline close on both aggregate and cross-sectional comovements

Decomposition of consumption response

Details

Shapley values of each channel

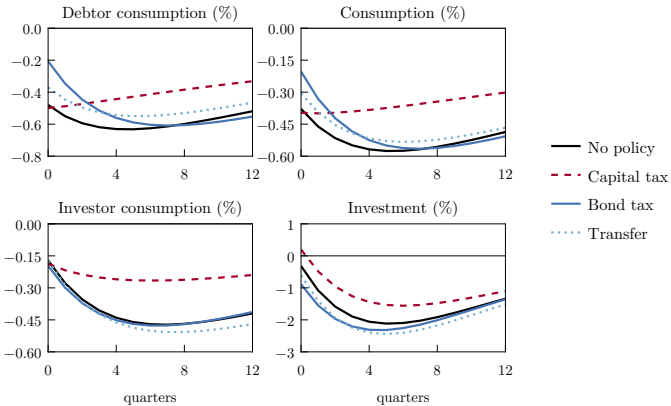
- Decompose each channel using tax wedges $(1 - \tau_t^{Rj})R_{t+1}^j$ undoing premia changes



- Credit supply contributes most of the consumption response over first the first year
- Higher safe rates offset capital channels in short-term

Countercyclical macroprudential policy

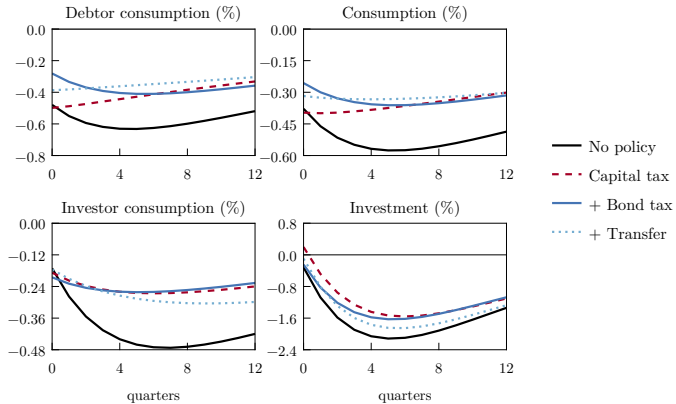
- Taxes on bond and capital returns, rising with each premium



- Bond tax: smaller initial fall, but crowds out investment
- Capital tax: faster recovery, but no support on impact

Countercyclical macroprudential policy

- Taxes on bond and capital returns, rising with each premium



- Both together: smaller initial fall and faster recovery
- Bond tax far cheaper than transfers

Conclusion

- Key Q in macro-finance: how do risky asset prices affect real activity?
- This paper:
  - Households with consumer credit are most responsive to risk prices
  - Credit supply accounts for most of the aggregate consumption response
  - Stabilisation policies in different markets operate at different frequencies
- Open questions: drivers of risk premia, measuring the household credit surface

Leverage as a risk aversion shock

Back

- A fraction ϵ of investors run intermediaries
- In a morning market, intermediaries issue equity to the other investors
- A fraction $\theta_0 = \theta \exp(\epsilon_0^\theta)$ then draw an investment opportunity
- The other $1 - \theta_0$ lend deposits at the risk-free rate i_0
- Shock ϵ_0^θ changes leverage within the intermediary sector
- Equilibrium risk premia

$$\mathrm{rp}_0^b = \sigma_b^2 \frac{1}{\theta_0} \frac{-a_0^d}{a_0^i}, \qquad \mathrm{rp}_0^k = \sigma_z^2 \frac{1}{\theta_0} \frac{a_0^i + a_0^d}{a_0^i}.$$

- Equivalent to baseline with $\gamma_0 = 1/\theta_0$. Equilibrium otherwise unchanged.

Full equilibrium definition

Back

Given states $\{\eta_t, \eta_t^f, w_{t-1}, k_{t-1}, \gamma_t\}$ and shock processes $\{\epsilon_t^z, \epsilon_t^\gamma\}$, an equilibrium is a set of prices and policies such that

- Households maximise utility s.t. budget and portfolio constraints
- Intermediary maximises objective, firm maximises profits, union maximises welfare
- Taylor rule holds; foreign rate satisfies $R_t^f = (1 + r^\star)\left(1 - \frac{1-\omega}{\omega} f_t\right)$
- Markets clear

$$\text{Goods: } \lambda^i c_t^i + \lambda^d c_t^d + (k_t/k_{t-1})^{\chi^k} x_t = y_t, \qquad \text{Capital: } k_t = \lambda^i k_t^i + \lambda^d k_t^d,$$

$$\text{Risky bonds: } 0 = \lambda^i b_t^i + \lambda^d b_t^d, \qquad \text{Safe bonds: } 0 = \lambda^i m_t^i + \lambda^d m_t^d$$

- Investor and foreign wealth shares evolve as

$$\eta_{t+1} = (1-\xi)\,\frac{\lambda^i\left[(\Pi_{t+1} + (1-\delta)Q_{t+1}^k)k_t^i + R_{t+1}^{b,0} b_t^i + R_t^f b_t^f\right]}{(\Pi_{t+1} + (1-\delta)Q_{t+1}^k)k_t} + \xi\bar{\eta},$$

$$\eta_{t+1}^f = (1-\xi)\,\frac{-\lambda^i b_t^f}{(\Pi_{t+1} + (1-\delta)Q_{t+1}^k)k_t}$$

Group sizes in the CEX

Back

	Households	Months	Households per month	Mean share (%)
Cons. credit & securities	13,103	375	35	10.5
Cons. credit, no securities	59,109	375	158	47.9
Securities, no cons. credit	7,010	375	19	5.9
Neither	43,185	375	115	35.7

Consumption, including durables

Back

	Consumption	
	(1)	(2)
EBP	-0.137 (0.904)	—
Sec., no cons. credit × EBP	-0.454 (1.968)	-0.592 (2.256)
Cons. credit, no sec. × EBP	-2.207*** (0.763)	-2.965*** (1.068)
Cons. credit & sec. × EBP	-6.860*** (1.790)	-7.476*** (2.020)
Fixed effects	None	Demog. × date
Observations	119,304	119,304

Bekaert, Engstrom and Xu (2022) Back

	Nondurables and services	
	(1)	(2)
Risk aversion	-0.012 (0.040)	—
Sec., no cons. credit × Risk aversion	-0.085** (0.039)	-0.070* (0.040)
Cons. credit, no sec. × Risk aversion	-0.076*** (0.021)	-0.062** (0.024)
Cons. credit & sec. × Risk aversion	-0.118** (0.046)	-0.111*** (0.038)
Fixed effects	None	Demog. × date
Observations	122,407	122,407

Excess CAPE yield

Back

	Nondurables and services	
	(1)	(2)
Excess CAPE yield	-0.132 (0.553)	—
Sec., no cons. credit × Excess CAPE yield	-0.522 (0.928)	-0.354 (0.856)
Cons. credit, no sec. × Excess CAPE yield	-1.365** (0.528)	-1.220** (0.535)
Cons. credit & sec. × Excess CAPE yield	-1.236 (0.906)	-1.173 (0.844)
Fixed effects	None	Demog. × date
Observations	122,407	122,407

EBP in the CEX

Excluding NBER recession months

Back

	Consumption			Nondurables and servic	
	Full (1)	Excl. rec., 2020–21 (2)	Excl. rec., 2020–21 (3)	Full (4)	Excl. rec., 2020–21 (5)
EBP	-0.137 (0.904)	0.625 (2.032)	—	0.225 (0.774)	0.818 (1.603)
Sec., no cons. credit × EBP	-0.454 (1.968)	4.671 (3.623)	2.883 (4.558)	-1.001 (0.901)	1.549 (1.422)
Cons. credit, no sec. × EBP	-2.207*** (0.763)	-2.882 (1.905)	-4.805** (2.167)	-1.746*** (0.515)	-1.015 (1.066)
Cons. credit & sec. × EBP	-6.860*** (1.790)	-3.889 (3.371)	-5.703 (4.161)	-2.084 (1.271)	0.399 (1.399)
Fixed effects	None	None	Demog. × date	None	None

EBP in the CEX

Back

Joint estimation with mortgages

	Consumption	Nondurables and services
	(1)	(2)
EBP	0.310 (1.015)	0.199 (0.781)
Sec. × EBP	-3.231** (1.518)	-0.536 (0.860)
Consumer credit × EBP	-2.755*** (0.894)	-1.616*** (0.512)
Mortgage × EBP	-0.274 (1.125)	-0.114 (0.489)

Split by retirement accounts

Back

Heads under 65 years of age

	Nondurables and services	
	(1)	(2)
EBP	0.875 (0.828)	—
Sec., no cons. credit × EBP	-1.945 (1.426)	-2.365* (1.373)
Cons. credit, no sec., ret. × EBP	-2.905*** (0.608)	-3.441*** (0.718)
Cons. credit, no sec., no ret. × EBP	-1.752*** (0.548)	-2.307*** (0.533)
Cons. credit & sec. × EBP	-2.354* (1.240)	-3.046*** (1.032)
Fixed effects	None	Demog. × date
Observations	90,358	90,358

Food consumption

Back

	Food
	(1)
EBP	-1.831** (0.930)
Sec., no cons. credit × EBP	-0.169 (1.799)
Cons. credit, no sec. × EBP	-1.817** (0.718)
Cons. credit & sec. × EBP	0.187 (1.919)
Fixed effects	None
Observations	122,407

Consumption and wages

Back

	Nondurables and services	
	(1)	(2)
Wages	0.832 (0.541)	—
Sec., no cons. credit × Wages	-0.068 (0.741)	-0.156 (0.816)
Cons. credit, no sec. × Wages	0.177 (0.436)	0.218 (0.442)
Cons. credit & sec. × Wages	-0.645 (0.542)	-0.402 (0.625)
EBP	0.714 (0.729)	—
Sec., no cons. credit × EBP	-1.215 (1.005)	-0.755 (0.951)
Cons. credit, no sec. × EBP	-1.735*** (0.554)	-1.684*** (0.513)
Cons. credit & sec. × EBP	-2.510* (1.332)	-2.296** (1.069)
Fixed effects	None	Demog. × date
Observations	122,407	122,407

- Labour income does not replicate the ranking from the EBP

Consumption and unemployment

Back

	Nondurables and services	
	(1)	(2)
Unemployment	-2.283*** (0.431)	—
Sec., no cons. credit × Unemployment	-1.670*** (0.505)	-1.266** (0.576)
Cons. credit, no sec. × Unemployment	-0.086 (0.466)	0.109 (0.501)
Cons. credit & sec. × Unemployment	-2.061** (0.837)	-1.878** (0.861)
EBP	1.240 (0.762)	—
Sec., no cons. credit × EBP	-0.402 (0.974)	-0.090 (0.896)
Cons. credit, no sec. × EBP	-1.819*** (0.520)	-1.796*** (0.503)
Cons. credit & sec. × EBP	-1.563 (1.143)	-1.520* (0.905)
Fixed effects	None	Demog. × date
Observations	122,407	122,407

Consumption and TIPS

Back

	Nondurables and services	
	(1)	(2)
5y real rate	0.151 (0.945)	—
Sec., no cons. credit × 5y real rate	-0.197 (1.580)	-0.335 (1.509)
Cons. credit, no sec. × 5y real rate	-0.084 (0.681)	-0.134 (0.779)
Cons. credit & sec. × 5y real rate	1.121 (1.765)	1.719 (1.979)
EBP	-1.011 (1.060)	—
Sec., no cons. credit × EBP	-1.821** (0.833)	-0.927 (0.830)
Cons. credit, no sec. × EBP	-2.055*** (0.598)	-1.507** (0.611)
Cons. credit & sec. × EBP	-3.780*** (1.244)	-3.179** (1.325)
Fixed effects	None	Demog. × date
Observations	81,148	81,148

Calibration: Portfolios in the SCF CEX Quant model

- Debtors: hh with consumer credit, investors: liquid capital, no consumer credit

	Share of total (%)			Share of capital (%)		
	Population	Labor income	Total income	Debt	Liquid capital	Illiquid capital
Debtors	61.4	59.5	55.6	1.7	20.7	8.6
Investors	15.3	27.1	32.6	0.0	58.1	10.5
Residual	23.3	13.4	11.8	0.0	0.0	2.2

- Debt: mortgages, credit cards (revolving), auto loans, other install. loans
- Liquid capital: equities held directly or through funds, private business wealth
  - Adjusted for leverage using Financial Accounts
- Illiquid capital: equities in ret. accounts, vehicles, houses, other non-fin. assets

External calibration

Back

Description	Value	Notes
ξ: death rate	0.01	
λ^i: investor population mass	0.199	SCF
λ^d: debtor population mass	0.801	SCF
$\bar{l}^i$: investor labor allocation	0.31	SCF
$\bar{l}^d$: debtor labor allocation	0.69	SCF
$\bar{\eta}$: newborn investor wealth share	0.199	λ^i
ν: Frisch elasticity	0.75	Chetty et al. (2011)
α: capital share	0.36	
δ: depreciation rate	0.025	
ε: elast. subs. workers	5	
χ^W: Rotemberg costs	663	Grigsby et al. (2021)
χ^k: capital adjustment exponent	2	
ϕ: Taylor coefficient	1.5	Taylor (1993)
p^d: mean disaster probability	0.5%	Barro (2006)
φ^z: productivity disaster	10%	Nakamura et al. (2013)
ρ^γ: intermediary CRRA persistence	0.8	$\rho(EBP_t, EBP_{t-1})$

Internal calibration

Back

Description	Value	Moment	Target	Model
β^i: discount factor	0.988	4E[risk-free rate]	0.7%	0.7%
β^d: discount factor	0.966	E[η^i]	72.4%	72.4%
γ: intermediary risk aversion	34.338	4E[eq. prem.]	7.9%	7.9%
$\bar{i}$: monetary policy intercept	0.002	4E[infl.]	0.0%	0.0%
σ_z: productivity volatility	0.004	$\sigma(\Delta \log I)/\sigma(\Delta \log y)$	1.2	1.2
ψ: IES	0.774	$\sigma(\Delta \log x)$	1.8%	1.8%
$\bar{k}$: min. capital	0.155	E[debtor cap./total cap.]	29.3%	29.3%
σ_γ: intermediary risk aversion vol.	1.775	$\sigma(\Delta \log c)$	0.6%	0.6%
z^b: bond default intercept	-0.137	4E[charge-off rate]	4.5%	4.5%
κ: bond default slope	32.82	$\sigma(\Delta$ charge-off rate)	0.6%	0.6%
φ^b: bond default disaster	0.132	4E[bond prem.]	6.8%	6.8%

Internal calibration

Back

Representative agent

Description	Value	Moment	Target	Model
β^i: discount factor	0.985	4E[risk-free rate]	0.7%	0.7%
γ: intermediary risk aversion	34.473	4E[eq. prem.]	7.9%	7.9%
$\bar{i}$: monetary policy intercept	0.002	4E[infl.]	0.0%	0.0%
σ_z: productivity volatility	0.004	$\sigma(\Delta \log I)/\sigma(\Delta \log y)$	1.2	1.2
ψ: IES	0.680	$\sigma(\Delta \log x)$	1.8%	1.8%
σ_γ: intermediary risk aversion vol.	2.030	$\sigma(\Delta \log c)$	0.6%	0.6%

Decomposing transmission channels using tax wedges

Back

- Intermediary prices each asset as if its return were $(1 - \tau_t^{Rj})R_{t+1}^j$ along IRF
- Household credit channel: undo change in $E[R_{t+1}^b - R_{t+1}^d]$

$$E_t^{\{\tau\}}\left[R_{t+1}^b - R_{t+1}^d\right] = E_t\left[R_{t+1}^b - R_{t+1}^d \mid \gamma = \bar{\gamma}\right]$$

- Capital channels: undo change in $E[R_{t+1}^k - R_{t+1}^d]$

$$E_t^{\{\tau\}}\left[R_{t+1}^k - R_{t+1}^d\right] = E_t\left[R_{t+1}^k - R_{t+1}^d \mid \gamma = \bar{\gamma}\right]$$

- Contribution: Shapley values of the surviving response, C^0 is the no-shock path

$$\Delta C^{\{\tau\}} = \sum_t \left(C_t^{\{\tau\}} - C_t^0\right).$$